Zach Stoebner

zstoebner@austin.utexas.edu | zstoebs.github.io

Overview

PhD researcher working on provably efficient, scalable, and robust algorithms for large-scale inverse problems, with a focus on **machine learning**, **optimization**, and **probability**.

Education

University of Texas at Austin *Austin, TX*

PhD in Electrical & Computer Engineering (advised by Prof. Jon Tamir) *2028 (Expected)*

- Topic: Theory and algorithms for operator learning in large-scale inverse problems
- Coursework: Convex Optimization, Probability & Stochastic Processes, Real Analysis, Large-Scale Optimization, Advanced Probability, Tensor Methods

Vanderbilt University *Nashville, TN*

BS in Computer Science & Neuroscience, Minor in Applied Mathematics *2021*

MS in Computer Science (advised by Prof. Ipek Oguz) *2022*

- Thesis: Deep learning for video segmentation
- Coursework: Statistical ML, Systems Theory, Computational Game Theory, Intelligent Systems & Robotics

Work

Adobe *Bay Area, CA*

AI/ML Research Intern, Firefly *Summer 2025*

- Developed state-space models for embedding compression in long-form text-to-image generation.
- Won the Firefly intern research showcase.

Research

Publications

[1] "Randomized variable-metric monotone operator learning for linear inverse problems", **ZS**, Tao Hong, Jonathan I. Tamir. (*Manuscript in preparation*) [Poster]

- Developed a variable-metric forward-backward splitting framework for monotone operator learning.
- Designed a low-rank second-order method using randomized Nyström approximation.
- Proved high-probability stability, robustness, and contraction guarantees via random matrix theory.

[2] "INFusion: diffusion regularized implicit neural representations for 2D and 3D accelerated MRI reconstruction", Yamin Arefeen, Brett Levac, **ZS**, Jonathan I Tamir. *IEEE-Asilomar* (2024). [DOI][arXiv]

[3] "Reducing malware analysis overhead with coverings", Michael Sandborn, **ZS**, Westley Weimer, Stephanie Forrest, Ryan Dougherty, Jules White, Kevin Leach. *IEEE-TDSC* (2023). [DOI][Code]

[4] "Comprehensive shape analysis of the cortex in Huntington's disease", **ZS**, Kilian Hett, Ilwoo Lyu, Hans Johnson, Jane S. Paulsen, Jeffrey Long, Ipek Oguz. *Human Brain Mapping* (2023). [DOI][Code]

[5] "Segmentation of kidney stones in endoscopic video feeds", **ZS**, Daiwei Lu, Seok Hee Hong, Nicholas L. Kavoussi, and Ipek Oguz. *SPIE Medical Imaging: Image Processing* (2022). [DOI][arXiv]

Posters

"Jointly solving for system identification and signal reconstruction with implicit neural representations", **ZS**, Jonathan I. Tamir. *{6G@UT, Oden SciML Workshop}* (2024) [Code][Poster]

Patents

"Systems and Methods for Navigation and Identification during Endoscopic Surgery", Nicholas L. Kavoussi, Ipek Oguz, Jie Ying Wu, **ZS**, Daiwei Lu, Ayberk Acar.
U.S. Patent Application No. 18/622,134.

Accolades

2026-2027 **Tolbert Endowed Fellowship**, University of Texas at Austin

2022-2026 **Cockrell Engineering Doctoral Fellowship**, University of Texas at Austin

2021 **Research Honors in Computer Science**, Vanderbilt University

2017-2021 **Dean's List**, Vanderbilt University

Skills

Programming: Python, C++, C

Tools: PyTorch, Git, Linux

Languages: English (native), Portuguese (fluent), Spanish (advanced), French (basic)